Principles of Induction

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1 Induction on the set of Natural Numbers

For this section, we denote the set of natural numbers by the familiar $\mathcal{N}$, and assume that $\mathcal{N}$ begins with 1 (not 0).

Theorem 1.1. The following are equivalent.

Principle of (Weak) Induction:
(*Basis*) $P(1)$ is true
(*Induction Step*):
For all $n \in \mathcal{N}$, if $P(n)$ is true then $P(n+1)$ is true
For all $n \in \mathcal{N}$, $P(n)$ is true.

Principle of Strong Induction:
(*Basis*) $P(1)$ is true
(*Induction Step*):
For all $n \in \mathcal{N}$, if $P(i)$ is true for all $i = 1, \ldots, n$ then $P(n+1)$ is true
For all $n \in \mathcal{N}$, $P(n)$ is true.

Proof. ($1 \Rightarrow 2$): To prove 2, assume the Basis and Induction Step of the Principle of Strong Induction for a statement $P(n)$. Now, for any $n \in \mathcal{N}$, consider the statement

$$Q(n) : P(i) \text{ is true for all } i = 1, \ldots, n.$$

Then the Basis and Induction Step of 1 hold for $Q(n)$, whereby, by 1, for all n, $Q(n)$ is true, that is $P(n)$ is true for all n.

$(2 \Rightarrow 1)$: Easy. $\square$